

Reviewer Pack — Minimal Rank of Noncrossed Products Bounds BP Read-Twice Wid...

Entry 3e215381ba21 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-28 23:03:13 UTC

Minimal Rank of Noncrossed Products Bounds BP Read-Twice Width

Entry ID: 3e215381ba21 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 23:03:13 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry (Noncrossed Products) **Field B** (complexity object): Branching Program: Read-twice Complexity

Statement:

[‘For a given instance of the MAX-CUT problem, let P be its corresponding read-twice BP. The minimal rank of the noncrossed product algebra associated with the adjacency matrix of the graph in the instance is polynomially bounded by the width of the BP P , i.e., $E[\text{rank}(\text{noncrossed_product_algebra})] = O(\text{width}(P))$ for all instances with $n \leq 40$.’, ‘Further, for a random instance of MAX-CUT with n variables, if the minimal rank of the noncrossed product algebra is greater than a certain threshold T that depends on n , then the BP width of P is at least $\Omega(n^{1/2})$.’]

Rationale (proposer’s reasoning):

[“The noncrossed product algebra captures information about the group structure of the automorphism group of the MAX-CUT graph. By studying its rank, we may expose a deep connection between the underlying graph’s properties and the complexity of representing it as a read-twice BP.”, “This conjecture, if true, would provide a new invariant for BP width that could be used to distinguish between different complexity classes.”]

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 320dd3f78147b8f1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all instances with $n \leq 40$, $E[\text{rank}(\text{noncrossed_product_algebra})]$ is within a factor of 3 from $O(\text{width}(P))$ as calculated across at least 30 seeds, and falsified if any seed produces a discrepancy greater than 3 or the BP width is less than $\Omega(n^{1/2})$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=90.8s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(matrix, b):
    n = len(matrix)
    augmented = [[matrix[i][j] for j in range(n)] + [b[i]] for i in range(n)]

    for i in range(n):
        # Find pivot row
        max_row = i
        for k in range(i+1, n):
            if abs(augmented[k][i]) > abs(augmented[max_row][i]):
                max_row = k

        # Swap rows
        augmented[i], augmented[max_row] = augmented[max_row], augmented[i]

        # Eliminate below pivot
        for k in range(i+1, n):
            factor = Fraction(augmented[k][i], augmented[i][i])
            for j in range(n + 1):
                augmented[k][j] -= factor * augmented[i][j]

    # Back substitution
    x = [0] * n
    for i in range(n-1, -1, -1):
        x[i] = augmented[i][-1]
        for k in range(i+1, n):
            x[i] -= augmented[i][k] * x[k]
        x[i] /= augmented[i][i]

    return x
```

```

def adjacency_matrix(graph):
    n = len(graph)
    mat = [[0] * n for _ in range(n)]
    for u, v in graph:
        mat[u][v] = 1
        mat[v][u] = 1
    return mat

def laplacian_matrix(mat):
    n = len(mat)
    lmat = [[0] * n for _ in range(n)]
    for i in range(n):
        degree = sum(mat[i])
        lmat[i][i] = degree
        for j in range(i+1, n):
            lmat[i][j] = -mat[i][j]
            lmat[j][i] = -mat[i][j]
    return lmat

def bp_width(instance):
    graph = instance['graph']
    n = len(graph)
    mat = adjacency_matrix(graph)
    laplacian = laplacian_matrix(mat)

    # Compute eigenvalues using Gaussian elimination
    try:
        eigenvalues = gaussian_elimination(laplacian, [0] * n)
    except ZeroDivisionError:
        return None

    # The width of the BP is the maximum absolute value of the eigenvalues
    bp_w = max(abs(e) for e in eigenvalues if e != 0)
    return bp_w

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    graph = [(i, j) for i in range(n) for j in range(i+1, n)]
    instance = {'graph': graph}

    bp_w = bp_width(instance)
    if bp_w is None:
        return {
            'metric_name': 'bp_width',
            'metric_value': None,
            'instances_tested': 1,
            'conjecture_holds': False,
            'counterexample': 'mapping_undefined'
        }

    min_rank = len(graph) # Simplified for testing purposes

    return {

```

```

        'metric_name': 'bp_width',
        'metric_value': bp_w,
        'instances_tested': 1,
        'conjecture_holds': False,
        'counterexample': ''
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r['metric_value'] for r in results if r['metric_value'] is not None]
    support_fraction = sum(r['conjecture_holds'] for r in results) / len(results)

    if all(v is not None for v in metric_values):
        mean = sum(metric_values) / len(metric_values)
        std = (sum((x - mean) ** 2 for x in metric_values) / len(metric_values)) ** 0.5
        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")
        else:
            print(f"RESULT: FALSIFIED counterexample='' first_failing_seed=None")
    else:
        print("RESULT: INCONCLUSIVE reason=missing_data n_tested=30")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'mapping_
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
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TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
TRIAL: {'metric_name': 'bp_width', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': 
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_952854f2.py", line 124, in <module>
    mean = sum(metric_values) / len(metric_values)
    ~~~~~^~~~~~
ZeroDivisionError: division by zero

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which means that the conjecture could not be evaluated according to the pre-registered support condition. | next: Re-run the test with proper error handling and ensure it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12507
2	preregistration	ollama_remote	glm4:latest	0	0	6689
3	novelty	ollama_remote	glm4:latest	0	0	4908
4	novelty	ollama_remote	glm4:latest	0	0	5117
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	43074
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12968
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13358
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12594
9	judge	ollama_remote	glm4:latest	0	0	13588

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 124802 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/3e215381ba21.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/3e215381ba21.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/3e215381ba21.tar.gz (if generated)